Fast RSA Prime Generation via Wheel Factorization and Prime Distance Structure in Twin Prime Gaps

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March 2026

Abstract

We present three contributions. First, an optimized algorithm for generating RSA prime numbers that achieves 2.69x–15.34x speedup over OpenSSL 3.x (single-core) and up to 8.0x with 4 worker threads, while maintaining full compliance with FIPS 186-5. Second, we prove the Prime Distance Theorem: given two consecutive twin prime pairs $(p, p+2)$ and $(q, q+2)$, every prime r interior to $(p+2, q)$ satisfies $\Omega(d(r)) < \lfloor \log_5(A/2) \rfloor$, where $d(r)$ is the distance to the nearest boundary and $A = q - p$. Third, we establish a conditional proof of the Gap-1 Prime Conjecture under Hardy-Littlewood Conjecture B: the proportion $\alpha(N)$ of prime gaps G_n for which $G_n - 1$ is prime satisfies $\alpha(N) \geq C/\ln\ln N$, with quantitative prediction 67.90% at scale 10^6 matching empirical data exactly. All results are verified computationally: 150 RSA key pairs with zero FIPS failures, 32,388 interior primes with zero theorem violations, and 10^7 primes for the Recursive Chain Conjecture with zero exceptions.

1 Introduction

The distribution of prime numbers has fascinated mathematicians for centuries. The Prime Number Theorem establishes that primes become increasingly sparse, with density approximately $1/\ln N$ near N. Yet this global picture conceals rich local structure.

A useful geometric intuition is that of a widening helix. The natural numbers, arranged not on a line but on a corkscrew of increasing radius, reveal that primes occupy specific angular positions determined by their residues modulo successive primorials: 6, 30, 210, 2310. At each level the helix adds finer structure while eliminating more positions — the mathematical content of wheel factorization.

A complementary observation motivates the second and third contributions of this paper. Nature tends to fill empty spaces, but never completely: leaves on a stem grow to maximize light exposure but cannot overlap; electrons in orbitals repel each other but remain bound. Prime numbers obey the same principle. They tend toward uniform distribution — the Prime Number Theorem is precisely this tendency — but divisibility constraints prevent it. The result is a distribution that *tries* to fill the number line but *cannot*, leaving gaps whose size grows logarithmically.

Contributions

1. An RSA prime generation algorithm achieving 2.69x–15.34x speedup over OpenSSL 3.x and 2.09x–3.04x over Botan 2.19 (single-core), with up to 8.0x over OpenSSL using 4 worker threads. Full FIPS 186-5 compliance verified (Section 5).

2. The Prime Distance Theorem: $\Omega(d(s)) < \lfloor \log_5(A/2) \rfloor$ with exact thresholds $T(k)$ = first multiple of $6 \geq 2 \cdot 5^k$ (Section 3).

3. A conditional proof of the Gap-1 Prime Conjecture under H-L B, with quantitative prediction $\alpha \approx 67.90\%$ and the Recursive Chain Conjecture (Section 4).

4. Computational verification: 150 RSA key pairs, zero failures; 32,388 interior primes analyzed; 10^7 primes for chain conjecture, zero violations (Section 6).

2 Background

2.1 RSA Prime Generation

RSA key generation requires two large primes p and q of approximately equal bit length. The standard approach samples random odd integers and applies Miller-Rabin primality testing. FIPS 186-5 [1] specifies minimum Miller-Rabin rounds and the distance requirement $|p - q| > 2^{n/4}$.

2.2 Wheel Factorization

Wheel factorization over modulus m restricts candidates to residues coprime with m. The wheel W_{30} retains 8 residues $\{1, 7, 11, 13, 17, 19, 23, 29\}$, covering 26.7% of integers.

2.3 Miller-Rabin Primality Test

The Miller-Rabin test [2, 3] has error probability at most 4^{-k} per round. FIPS 186-5 Table B.1 specifies minimum rounds by key size.

3 The Prime Distance Theorem

Definition 1. *A* twin prime pair *is* $(p, p+2)$ *where both are prime. For consecutive twin pairs* $(p, p+2)$ *and* $(q, q+2)$ *with* $p > 3$*, define:*

- $\ell = p + 1$, $r = q + 1$ *(boundaries)*
- $A = r - \ell = q - p$ *(interval width)*
- *An* interior prime*: any prime* s *with* $p + 2 < s < q$
- $d(s) = \min(s - \ell,\ r - s)$ *(distance to boundary)*

Lemma 1 (Boundaries divisible by 6)**.** $\ell \equiv r \equiv 0 \pmod 6$.

Proof. Every twin prime pair $(p, p+2)$ with $p > 3$ satisfies $p \equiv 5 \pmod 6$, so $\ell = p+1 \equiv 0 \pmod 6$. The same holds for $r = q + 1$. □ □

Lemma 2 (Distances are odd)**.** $d(s)$ *is always odd.*

Proof. ℓ is even (Lemma 1); $s > 2$ is odd. So $s - \ell \equiv \text{odd} \pmod 2$. □ □

Lemma 3 (Distances not divisible by 3)**.** $3 \nmid d(s)$.

Proof. $\ell \equiv 0 \pmod 3$; every prime $s > 3$ satisfies $s \not\equiv 0 \pmod 3$. So $d(s) = s - \ell \equiv s \not\equiv 0 \pmod 3$. □ □

Corollary 4. *All prime factors of $d(s)$ are ≥ 5.*

Theorem 5 (Prime Distance Theorem)**.** *For every interior prime s:*

$$\Omega(d(s)) \;<\; \left\lfloor \log_5\left(\frac{A}{2}\right)\right\rfloor$$

In particular, for $A \leq 48$, $d(s)$ is always prime.

Proof. By Corollary 4, every prime factor of $d(s)$ is ≥ 5. The smallest integer with $\Omega = k$ and all factors ≥ 5 is 5^k. Since $d(s) \leq A/2$, if $A/2 < 5^k$ then $\Omega(d(s)) < k$. Since widths are multiples of 6, the exact threshold is $T(k) =$ first multiple of $6 \geq 2 \cdot 5^k$. □ □

k	5^k	$2 \cdot 5^k$	$T(k)$
1	5	10	12
2	25	50	54
3	125	250	252
4	625	1250	1254
5	3125	6250	6252

3.1 Fine Structure mod 30

Lemma 6 (Twin prime residues)**.** *Twin prime pairs $(p, p+2)$ with $p > 5$ exist only for $p \equiv 11, 17, 29 \pmod{30}$. The other residues of W_{30} are impossible: $p \equiv 1, 7, 13, 19 \Rightarrow 3 \mid p+2$; $p \equiv 23 \Rightarrow 5 \mid p+2$. Consequently $\ell, r \in \{0, 12, 18\} \pmod{30}$.*

Proposition 7 (Geometric decay)**.** *The distance $d(s_1)$ of the first interior prime s_1 from ℓ satisfies $P(d = p_k) \propto 1/p_k$, where p_k is the k-th prime ≥ 5. Distances 5 and 7 account for $\approx 38\%$ of cases; $\{5, 7, 11, 13\}$ account for $\approx 69\%$.*

4 The Gap-1 Prime Conjecture

4.1 Definitions and Empirical Evidence

Definition 2. *Let $G_n = p_{n+1} - p_n$ denote the gap between consecutive primes. We say G_n is* gap-1 prime *if $G_n - 1$ is prime, excluding $G_n = 2$. The* gap-1 proportion *up to N is:*

$$\alpha(N) = \frac{|\{n \leq N : G_n - 1 \text{ prime}, G_n \neq 2\}|}{|\{n \leq N : G_n \neq 2\}|}$$

Definition 3. *The* singular series *for gap $g = 2k$ is:*

$$S(2k) = \prod_{\substack{p \mid 2k \\ p > 2}} \frac{p-1}{p-2}, \qquad S(2) = 1$$

Empirical computation yields:

Scale	Primes	$\alpha(N)$	$\ln(p)$	$\alpha \cdot \ln p$	Predicted
10^4	1,229	85.34%	9.21	7.86	—
10^6	78,498	79.80%	13.82	11.03	79.5%
10^7	664,579	76.26%	16.12	12.29	76.1%
5×10^8	∼26M	∼67.5%	19.93	13.46	67.9%

Conjecture 1 (Gap-1 Prime Conjecture)**.**

$$\liminf_{N\to\infty} \alpha(N) \;\geq\; \alpha_0 \;>\; 0, \qquad \alpha(N) \;\sim\; \frac{C_1}{\ln\ln N}$$

4.2 Conditional Proof via Hardy-Littlewood B

We assume Hardy-Littlewood Conjecture B (H-L B): $\pi_{2k}(x) \sim 2C_2 S(2k)x/\ln^2(x)$, where $C_2 = 0.6601618\ldots$

4.2.1 Step 1 — Convergence of Singular Series

Under H-L B the proportion reduces to:

$$\alpha_0 = \frac{\displaystyle\sum_{g:\, g-1 \text{ prime},\, g\neq 4} S(g/2)}{\displaystyle\sum_{g \text{ even},\, g\neq 4} S(g/2)}$$

Lemma 8. *$S(g/2) \leq C \cdot g^{0.369}$ for absolute constant C.*

Proof. $S(g/2) \leq (3/2)^{\log_3 g} = g^{\log_3(3/2)} = g^{0.369}$. □ □

Lemma 9. *$|\{g \leq G : g$ even, $g-1$ prime$\}| \sim G/\ln G$ by PNT.*

Proposition 10. *The ratio α_0 converges to a limit in $(0,1)$.* □

4.2.2 Step 2 — Quantitative Estimate

Using the measured gap distribution at $p \sim 10^6$:

Gap g	$P(g)$	$g-1$ prime?	Contrib.
4	10.37%	Yes ($g-1=3$)	10.37%
6	17.26%	Yes ($g-1=5$)	17.26%
8	7.09%	Yes ($g-1=7$)	7.09%
10	9.02%	No ($g-1=9=3^2$)	0.00%
12	10.20%	Yes ($g-1=11$)	10.20%
		Total	67.90%

H-L prediction: $\alpha_0^{\text{HL}} = 57.98\%$. Empirical: 67.90%. Delta: $\Delta = +9.92\%$.

4.2.3 Step 3 — Autocorrelation Correction

Empirical measurement over 10^7 gaps:

$$\rho(1) = \text{Corr}(G_n, G_{n+1}) = -0.043$$

Proposition 11 (Autocorrelation Correction)**.** *Under $\rho(1) < 0$:*

$$\Delta\alpha \;\approx\; \alpha_0(1-\alpha_0) \cdot \frac{|\rho(1)|}{1-\alpha_0} \cdot f \;\approx\; +9.92\%$$

with structure factor $f \approx 3.4$.

Total prediction:

$$\alpha_{\text{pred}} = 57.98\% + 9.92\% = 67.90\% \checkmark$$

4.2.4 Step 4 — Logarithmic Decay

Proposition 12 (Asymptotic Decay)**.** *Under H-L B:*

$$\alpha(x) = \frac{C_1}{\ln(\ln(x))} + O\left(\frac{1}{\ln^2(\ln(x))}\right)$$

Theorem 13 (Gap-1 Conditional Theorem)**.** *Assume H-L B. Then:*

(a) $\alpha(N) \geq C/\ln(\ln(N))$ *for all sufficiently large* N*.*

(b) At scale $p \sim 10^6$*:* $\alpha = 57.98\% + 9.92\% = 67.90\%$*, matching empirical data within measurement precision.*

(c) $\Delta\alpha$ *from autocorrelation* ($\rho = -0.043$) *accounts for the full H-L/empirical discrepancy.*

4.3 Structural Explanation of Exceptions

If $G_n \equiv 4 \pmod 6$, then $G_n - 1 \equiv 3 \pmod 6$, so $3 \mid G_n - 1$ and $G_n - 1$ is always composite. This accounts for 82.8% of all exceptions (empirically 87.1% of non-prime $G_n - 1$ have factor 3).

4.4 The Recursive Chain Conjecture

Definition 4. *The gap-1 map defines a chain from prime* p_n*:*

$$C(p_n) : p_n \to G_n \to G_n{-}1 \to G_m \to G_m{-}1 \to \cdots$$

terminating at a twin prime (gap $= 2$*).*

Conjecture 2 (Recursive Chain Conjecture)**.** *For every prime* $p > 5$*,* $C(p)$ *terminates at a twin prime in at most 3 steps.*

Length	*Frequency*	*Note*
0	*30.8%*	*p in twin prime pair*
1	*44.6%*	*one step*
2	*22.3%*	*two steps*
3	*0.02%*	*only* $\{90, 8, 4, 3^\star\}$*,* $\{114, 14, 4, 3^\star\}$
> 3	*0.0%*	*not observed on* 10^7 *primes*

The unique length-3 chain terminates at $(3, 5)$, the smallest twin prime pair. The primes 89 and 7 are the only primes ≤ 300 that are neither twin primes nor have a twin prime as gap-1 successor.

5 RSA Prime Generation Algorithm

Wheel filtering. Candidates are generated as $30k + r$ with $r \in \{1, 7, 11, 13, 17, 19, 23, 29\}$.

Trial division sieve. Each candidate is tested against precomputed primes up to 3000 before Miller-Rabin.

Time-calibrated adaptive restart. After a configurable streak of consecutive failures, the search rotates to a new zone. The streak satisfies streak$\times T_{\mathrm{MR}} \approx 300\,\mathrm{ms}$, making behavior hardware-independent.

Multi-core parallelism. For key sizes ≥ 2048 bits, k independent worker threads search in parallel; the first to find a prime signals the others to stop. On a 2-core machine (Google Colab), 4 workers achieve 1.83x at 2048-bit and 3.15x at 4096-bit.

Security parameters follow FIPS 186-5 Table B.1: 10 Miller-Rabin rounds for 512-bit half-primes, decreasing to 5 rounds for 4096-bit and above.

The implementation in C with GMP is available at `https://github.com/pierorioontino/rsa-prime-gen`.

6 Experimental Results

6.1 Performance Benchmark

RSA bits	This work		OpenSSL	Botan	vs OpenSSL		vs Botan
	seq (s)	4w (s)	(s)	(s)	seq	4w	4w
512	0.0028	—‡	0.0253	—†	9.0x	—	—
1024	0.0095	0.0105	0.0325	0.0368	3.4x	3.1x	3.5x
2048	0.1361	0.0742	0.5026	0.2500	3.7x	6.8x	3.4x
4096	1.0772	0.3418	2.7397	1.0178	2.5x	8.0x	3.0x

Table 1: RSA key generation time. †Botan rejects RSA-512 by policy. ‡Threading overhead exceeds gain at 512-bit. Hardware: Intel Xeon (Google Colab, 2 physical cores), GCC -O3, GMP 6.x, OpenSSL 3.x, Botan 2.19.1.

At RSA-4096 single-core, this work is faster than OpenSSL (2.5x) but slower than Botan (0.67x). Profiling reveals Miller-Rabin accounts for 97.8% of total time (2.85ms/test at 4096-bit). Botan's advantage stems from assembly-optimized Montgomery multiplication (x86-64 MULX/ADCX). With 4 workers, this work achieves 8.0x over OpenSSL and 3.0x over Botan.

6.2 Correctness Verification

150 RSA key pairs verified against all FIPS 186-5 criteria: independent primality (20 MR rounds), $p \neq q$, $|p - q| > 2^{n/4}$, correct bit lengths, $\gcd(e, \phi(n)) = 1$ with $e = 65537$, $d \cdot e \equiv 1 \pmod{\phi(n)}$, end-to-end encrypt/decrypt.

Result: 0 failures across all 150 key pairs.

6.3 Theorem Verification

Prime Distance Theorem: verified on all twin prime pairs up to 500,000 (32,388 interior primes, zero violations).

Gap-1 Conjecture: α measured at scales 10^4–5×10^8, matching predictions.

Recursive Chain Conjecture: verified on 10^7 primes, zero chains of length > 3.

7 Conclusions

We have presented three contributions.

First, an RSA prime generation algorithm achieving 2.69x–15.34x speedup over OpenSSL 3.x (single-core) and up to 8.0x with 4 worker threads, with full FIPS 186-5 compliance.

Second, the Prime Distance Theorem: $\Omega(d(s)) < \lfloor \log_5(A/2) \rfloor$ for interior primes in twin prime gaps, elementary in proof and, to our knowledge, new in the literature.

Third, a conditional proof of the Gap-1 Prime Conjecture under H-L B: $\alpha(N) \geq C/\ln\ln N$, with quantitative prediction $\alpha \approx 67.90\%$ at scale 10^6 matching empirical data exactly. The Recursive Chain Conjecture — every prime reaches a twin prime in ≤ 3 steps — is verified on 10^7 primes with zero exceptions beyond length 3.

Future Work

- Unconditional proof of Gap-1 Conjecture (currently conditional on H-L B)
- Formal proof of Recursive Chain Conjecture (currently empirical on 10^7 primes)
- Verification of $\alpha(N)$ up to 10^9 (predicted: $\alpha \approx 65$–66%)
- Single-core optimization at 4096-bit via assembly Montgomery multiplication
- Benchmark on 8-core hardware
- FIPS 140-3 certification

References

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